

FILTER DESIGN TEST CASE

Ladder Filter - Band 25 (1900 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Ladder Filter filter targeting Band 25 (1900 MHz) for Automotive V2X Communication. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 2738.1 MHz and bandwidth of 355.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Ladder Filter
- Frequency Band: Band 25 (1900 MHz)
- Application: Automotive V2X Communication
- Substrate: 36° YX LiTaO3
- Package: Flip-Chip BGA
- Design Tool: Sonnet EM Suite v16
- Design Center: Andover HQ Lab

This specification applies to all variants of the Ladder Filter design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2738.1 MHz
Bandwidth (3 dB):	355.9 MHz
Insertion Loss Target:	<= 1.2 dB
Return Loss Target:	>= 9.0 dB
Out-of-Band Rejection:	>= 42.0 dB
Isolation (Duplexer):	>= 20.0 dB
Q Factor:	3407
Temperature Coefficient:	-13.9 ppm/°C
Die Size:	1.9 x 2.3 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	W

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the Ladder Filter template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 2738.1 MHz, BW 355.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 8214 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 534 MHz offset >= 42.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 4.8 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-10-14
Devices Tested: 69
Insertion Loss (meas): 0.87 dB
Return Loss (meas): 10.9 dB
Rejection (meas): 44.6 dB
Isolation (meas): 24.6 dB
Q Factor (meas): 3407
Group Delay Variation: 2.1 ns
Power Handling: +27 dBm
Die Yield: 84.8%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Ladder Filter design demonstrated below-target performance for Band 25 (1900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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